

How Deep Learning can support the Steel wheel design: a practical case study

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In the automotive industry, vehicle wheels play a crucial role in ensuring overall vehicle safety. A failure or an inadequate wheel design can compromise not only the vehicle's stability and handling, but also the safety of its occupants, potentially leading to severe consequences.

One of the key aspects in wheel design is fatigue strength, namely the ability of the wheel to withstand repeated cyclic loads without developing damage. Ensuring adequate fatigue resistance is essential to guarantee that the wheel maintains its safety performance throughout the entire service life of the vehicle.

However, wheel design is not limited to safety requirements alone. Automotive manufacturers also impose a wide range of additional constraints that must be satisfied, such as weight reduction, vehicle handling performance, compatibility with braking systems, wheel covers and valves, aesthetic appearance, and, increasingly important, environmental sustainability, including the reduction of CO₂ emissions throughout the product life cycle.

As a result, wheel design becomes a complex optimization process in which multiple competing factors must be carefully balanced. In many cases, achieving the optimal solution requires a large number of design iterations, often exceeding fifty. Each iteration is typically supported by advanced structural analyses based on the Finite Element Method (FEM), making the overall optimization process computationally intensive and potentially requiring several weeks to complete.

The objective of this work is to accelerate this development process by drastically reducing design times and enabling optimal solutions to be achieved within minutes through the application of Deep Learning technologies.

The proposed methodology aims to optimize the wheel design process by leveraging the capabilities of neural networks and Deep Learning techniques to significantly reduce computational time while maintaining a good level of accuracy and guiding designers toward the most promising design direction. The approach is structured into two main phases: the training phase and the validation phase.

1. Training phase

The first phase involves training a neural network using data obtained from FEM structural analyses. In this stage, the wheel dimensional parameters (namely the geometrical dimensions defining the wheel shape) are used as input variables for the neural network. The outputs consist of the stress results in the most critical areas obtained from the FEM simulations to which the wheel is subjected, the weight and the raw material needed for its construction.

During the training process, the model learns the complex relationships between the geometric parameters and the mechanical responses (stress distributions) of the wheel by processing large amounts of data. This learning process enables the neural network to develop an understanding of the wheel's structural behavior, allowing it to generate reliable predictions without the need to perform the numerous iterative FEM calculations traditionally required during the design process.

2. Validation Phase

Once trained, the neural network is tested and validated through a verification process in which the predicted results are compared with those obtained using traditional FEM methods. The outputs generated by the neural network are evaluated to assess their consistency and reliability with respect to the FEM data, showing a strong correlation between the two approaches.

A major advantage of the proposed methodology is that, once the model has been trained, it is capable of generating predictions within minutes rather than requiring weeks to complete FEM iterations across dozens or even hundreds of design configurations. The neural network can rapidly explore the design space by generating and evaluating tens of thousands of input–output combinations in a very short time. This enables a far more efficient search for the optimal solution, allowing the design team to focus on more creative and innovative aspects of the development process instead of repetitive computational iterations.

The present work is not limited to the description of the proposed methodology, but also includes exploratory results that demonstrate the effectiveness of the developed model. In particular, a practical case study is presented in which the neural network is used to explore different wheel geometric configurations while simultaneously optimizing structural performance and reducing computational time.

The speed and reliability offered by Deep Learning therefore not only accelerate the design process, but also provide a significant competitive advantage. The ability to rapidly evaluate multiple design configurations and identify optimal solutions much more efficiently than traditional approaches enables companies to respond more quickly to market demands and reduce the time-to-market of the final product.